1. **Question on system frequency regulation**

a) Assume that there are n identical generators, each with capacity of 50 kW and availability of 0.98.

i) The system design standard requires that the probability of meeting P kW peak demand is equal or greater than 0.99. As a system design engineer, formulate a general optimization problem to determine the minimum number of the generators n_{min} that meet this standard (P is a parameter in the optimization). Base on the formulated optimization problem, find n_{min} when $P = 100$.

$$\min \ n \tag{1.1}$$

$$\text{s.t. } P \leq 50n \tag{1.2}$$

$$0.98^n + \sum_{k=1}^{n-1} (C_n^k 0.98^{n-k} * 0.02^k)\delta_k \geq 0.99 \tag{1.3}$$

$$n \in Z \tag{1.4}$$

where δ_k is defined as follows:

$$\delta_k = \begin{cases} 1, \text{ if } 50(n-k) \geq P \\ 0, \text{ else} \end{cases} \tag{1.5}$$

[4 marks]

- $n = 2$, the probability of meeting 100 kW peak demand is $0.98^2 = 0.9604$, which is small than 0.99.
- $n = 3$, the probability of meeting 100 kW peak demand is $0.98^3 + 3 \times 0.98^2 \times 0.02 = 0.9988$, which is greater than 0.99.

$n_{min} = 3$.

[2 mark]

ii) Assume that the coincidence factor of an infinite number of households is 0.444. The peak demand of each individual household is 5 kW and the reliability criterion requires that the probability of meeting peak demand is equal or greater than 94%. If, at present, the number of generators n = 2, calculate the number of household that can be supplied?

- $n = 2$, to meet 94% reliability requirement, the maximum peak demand that can be served is 100 kW.

$$\left(0.444 + \frac{1 - 0.444}{\sqrt{m}}\right) m \times 5 = 100 \tag{1.6}$$

$m = 37.388$, 37 households

[2 marks]

iii) Now, the area has a plan for new property development and the total household number is increased **to** 58. Calculate and determine if the current system still meets the reliability criterion. If not, how many more generators are needed?

- $m = 58$, to meet 94% reliability requirement, the maximum peak demand that can be served is:

$$\left(0.444 + \frac{1-0.444}{\sqrt{m}}\right) m \times 5 = 149.9 \tag{1.7}$$

Solve the optimization in i) with $P = 149.9$ and modify the reliability from 0.99 to 0.94, giving $n_{min} = 3$

1 more generator is needed.

[2 marks]

b) Battery storage systems can provide frequency response much faster than thermal generators, therefore providing a service called Dynamic Containment (DC) in the UK. DC reduces the need for the regular frequency response (R), following this rule:

$$\left(\frac{H}{f_0} - \text{DC} \times \frac{T_{\text{DC}}}{4 \times \Delta f_{\max}}\right) \times R \geq \frac{(P_L - \text{DC})^2 \times T_d}{4 \times \Delta f_{\max}} \tag{1.8}$$

Nomenclature:

H:	System level of inertia (MW-s)
R:	Amount of frequency response scheduled (MW)
DC:	Amount of DCscheduled (MW)
P_{L}:	Value of generation outage we want to secure against (MW)
f_0:	Nominal frequency of the power grid (Hz)
$\Delta f_{\max}$:	Maximum admissible frequency deviation (Hz)
T_{d}:	Delivery time of frequency response (s)
T_{DC}:	Delivery time of DC (s)

Assumptions:

- The amount of frequency response: $R = 2000$ MWs,
- The cost of frequency response: \$5/MW,
- The largest generation unit in the UK: 1.8 GW,
- Other parameters: $f_0 = 50$ Hz, $\Delta f_{\max} = 0.8$ Hz, $T_d = 10$ s, $T_{\text{DC}} = 0.5$ s.

i) Based on (1.1), comment on how H, R, DC, T_d, T_{DC}, P_L affect the frequency nadir (the largest frequency reduction).

Increasing H, R, DC decreases frequency nadir; Increasing T_d, T_{DC}, P_L increases frequency nadir.

[1 mark each - 2 marks]

ii) Calculate the minimum required inertia in the system to maintain the frequency nadir constraint when DC =200 MW and 300 MW respectively.

- DC = 200 MW, H = 201.56 GWs
- DC = 300 MW, H = 178.13 GWs

[2 marks each - 4 marks]

iii) Assume that the frequency response is optimally procured at the current price. Determine the economic value of H in terms of \$/GWs on maintaining the frequency nadir constraint, in the case of DC = 300 MW.

$$g = \left(\frac{H}{f_0} - \text{DC} \times \frac{T_{\text{DC}}}{4 \times \Delta f_{\max}}\right) \times R - \frac{(P_L - \text{DC})^2 \times T_d}{4 \times \Delta f_{\max}} \geq 0 \tag{1.9a}$$

$$\frac{\partial g}{\partial R} = \left(\frac{H}{f_0} - \text{DC} \times \frac{T_{\text{DC}}}{4 \times \Delta f_{\max}}\right) = 3.5 \tag{1.9b}$$

$$\frac{\partial g}{\partial H} = \frac{R}{f_0} = 0.04 \tag{1.9c}$$

$$\text{Price of } H = 5 \times 0.04/3.5 \times 10^3 = \$57.14/\text{GWs} \tag{1.9d}$$

[4 marks]

2. **Question on per unit system**

a) How are the base voltages for the primary and secondary sides of an ideal transformer chosen? What is the benefit of this method?

The base voltages across a transformer are typically chosen as the nominal voltages of the primary and secondary sides.

In this way, the ideal transformer disappears in the per-unit system and can be modelled by a simple impedance.

This is because:

$$V_1^{pu} = V_2^{pu} \tag{2.1}$$

$$I_1^{pu} = I_2^{pu} \tag{2.2}$$

$$Z_1^{pu} = Z_2^{pu} \tag{2.3}$$

[4 marks]

b) Consider the high-voltage transmission circuit in Figure 2.1. The current at the receiving end lags V_r by 45° and $V_r = 1\angle 0°$ p.u., $|V_s| = 1.3$ p.u..

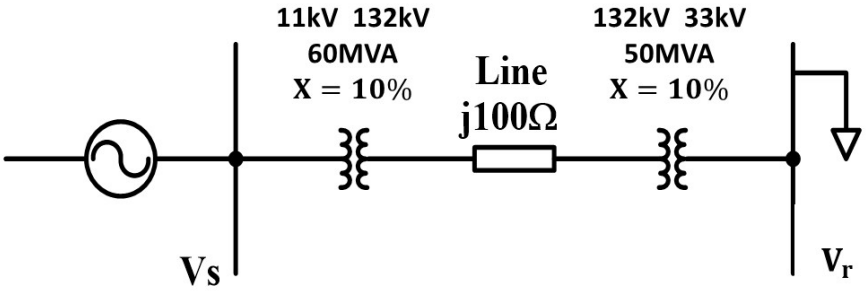


Figure 2.1 Transmission circuit with powers specified at both sending and receiving ends

i) Calculate the per unit values of the quantities (three impedances) in the figure, with properly selected bases.

Choose: $S_B = 50$ MW, $V_{B11} = 11$ kV, $V_{B132} = 132$ kV, $V_{B33} = 33$ kV.

[1 mark]

$$X_1 = \text{j}0.1\frac{50}{60} = 0.083\,\text{p.u.} \tag{2.4}$$

$$X_2 = \frac{\text{j}100}{V_{B132}^2/S_B} = 0.287\,\text{p.u.} \tag{2.5}$$

$$X_3 = \text{j}0.1\,\text{p.u.} \tag{2.6}$$

[3 marks]

ii) Draw the phasor diagram and calculate V_s and the magnitude of the line current in per unit.

Let $V_s = 1.3\angle\theta$.

Plot the phasor diagram.

[3 marks]

According to the diagram:

$$\frac{|V_s|\sin\theta}{|V_s|\cos\theta - |V_r|} = \tan(90° - 45°) \tag{2.7}$$

[2 marks]

$$1 = 1.3\cos\theta - 1.3\sin\theta \tag{2.8}$$

[2 marks]

$$\frac{1}{1.3^2} = 1 - 2\cos\theta\sin\theta \tag{2.9}$$

[1 mark]

$$\sin 2\theta = 1 - \frac{1}{1.3^2} = 0.408 \tag{2.10}$$

[2 marks]

$$\theta = 12.05^\circ \tag{2.11}$$

[1 mark]

$$I = \frac{1.3\sin\theta}{\sin 45^\circ X} = 0.8167\,\text{p.u.} \tag{2.12}$$

[1 mark]

3. **Question on power flow analysis and Gauss-Seidel method**

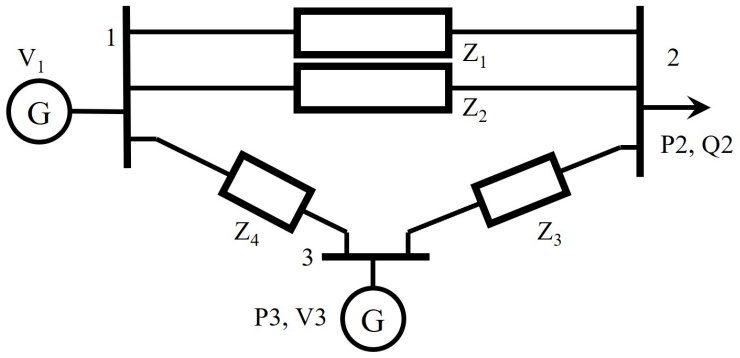


Figure 3.1 A three-bus power network

A three-bus power network is presented in Figure 3.1.

a) Express the bus admittance matrix for the system in terms of Z_1, Z_2, Z_3, Z_4.

Define $Z_0 = \frac{Z_1 Z_2}{Z_1+Z_2}$.

$$i_1 = \frac{1}{Z_0}(V_1 - V_2) + \frac{1}{Z_4}(V_1 - V_3) = \left(\frac{1}{Z_0} + \frac{1}{Z_4}\right)V_1 - \frac{1}{Z_0}V_2 - \frac{1}{Z_4}V_3 \quad (3.1)$$

$$i_2 = \frac{1}{Z_0}(V_2 - V_1) + \frac{1}{Z_3}(V_2 - V_3) = -\frac{1}{Z_0}V_1 + \left(\frac{1}{Z_0} + \frac{1}{Z_3}\right)V_2 - \frac{1}{Z_3}V_3 \quad (3.2)$$

$$i_3 = \frac{1}{Z_4}(V_3 - V_1) + \frac{1}{Z_3}(V_3 - V_2) = -\frac{1}{Z_4}V_1 - \frac{1}{Z_3}V_2 + \left(\frac{1}{Z_3} + \frac{1}{Z_4}\right)V_3 \quad (3.3)$$

[3 marks]

$$Y = \begin{bmatrix} \frac{1}{Z_0}+\frac{1}{Z_4} & -\frac{1}{Z_0} & -\frac{1}{Z_4} \\ -\frac{1}{Z_0} & \frac{1}{Z_0}+\frac{1}{Z_3} & -\frac{1}{Z_3} \\ -\frac{1}{Z_4} & -\frac{1}{Z_3} & \frac{1}{Z_4}+\frac{1}{Z_3} \end{bmatrix} \quad (3.4)$$

[2 mark]

b) When applying the Gauss-Seidel method to solve the load flow problem in a general multi-bus system, the bus voltages are updated in each iteration.

i) Derive the bus voltage update law for V_2 and V_3 in Figure 3.1 based on the node quantities given in the figure and the elements in the admittance matrix (assume V_1 is known). Explain why the bus voltages cannot be solved directly.

Based on $S = VI^*$ and $I = YV$, the voltage update law in a general multi-bus system can be derived as follows:

$$I_2 = \frac{S_2^*}{V_2^*} = \sum_{m\in\{1,\ldots,3\}} Y_{2m}V_m \quad (3.5)$$

$$V_2 = \frac{1}{Y_{22}}\left(\frac{S_2^*}{V_2^*} - \sum_{m\in\{1,\ldots,3\}\setminus 2} Y_{2m}V_m\right) \quad (3.6)$$

[1 mark]

$$I_3 = \frac{S_3^*}{V_3^*} = \sum_{m\in\{1,\ldots,3\}} Y_{3m}V_m \tag{3.7}$$

$$V_3 = \frac{1}{Y_{33}}\left(\frac{S_3^*}{V_3^*} - \sum_{m\in\{1,\ldots,3\}\setminus 3} Y_{3m}V_m\right) \tag{3.8}$$

[1 mark]

The bus voltages $\mathbf{V}$ can then be viewed as a nonlinear and implicit function of $\mathbf{V}$. As a result, a closed form of V cannot be obtained.

[1 mark]

ii) Based on the derived voltage update law, properly define the vector $\mathbf{x}$ and the multidimensional function $\mathbf{h}$, such that the Gauss-Seidel method can be generalized from the load flow problem to a nonlinear equation $\mathbf{x} = \mathbf{h}(\mathbf{x})$. Explain how to perform the Gauss-Seidel method in this general case.

For generalization, we define $\mathbf{x}$ as the bus voltages and $\mathbf{h}$ as the voltage update law. For an N-bus system, assuming voltage V_1 is known:

$$V_2 = h_2(V_1, V_2, \ldots V_N) \tag{3.9}$$
$$V_3 = h_3(V_1, V_2, \ldots V_N) \tag{3.10}$$
$$\ldots$$
$$V_N = h_N(V_1, V_2, \ldots V_N) \tag{3.11}$$

[1 marks]

- Choose an appropriate initial value for $\mathbf{x}^{(0)}$.
- Repeat the calculation of $\mathbf{x}^{(\nu+1)} = \mathbf{h}(\mathbf{x}^{(\nu)})$, $\nu = 0, 1, \ldots$
- If the sequence converges to $\mathbf{x}^*$, then $\mathbf{x}^*$ is a solution to the equation system $\mathbf{x} = \mathbf{h}(\mathbf{x})$, i.e., $\mathbf{x}^* = \mathbf{h}(\mathbf{x}^*)$
- Convergence Criterion: the algorithm stops if $\mathbf{x}^{(\nu)} - \mathbf{h}(\mathbf{x}^{(\nu)}) < \varepsilon$

[4 marks]

c) DC power flow approximation

i) The power flow between two buses can be written as: $P_{km} = \frac{V_k V_m \sin\theta_{km}}{X_{km}}$, $Q_{km} = \frac{V_k^2 - V_k V_m \cos\theta_{km}}{X_{km}}$, after neglecting line losses and shunt admittance. Comment on how the bus angle difference and voltage magnitude influence the active and reactive power respectively, by evaluating the sensitivities of P_{km} and Q_{km} with respective to θ_{km} and V_k at $\theta_{km} = 0$ (four quantities in total).

$$\frac{\partial P_{km}}{\theta_{km}} = \frac{V_k V_m}{X_{km}} \tag{3.12}$$

$$\frac{\partial P_{km}}{V_k} = 0 \tag{3.13}$$

$$\frac{\partial Q_{km}}{\theta_{km}} = 0 \tag{3.14}$$

$$\frac{\partial Q_{km}}{V_k} = \frac{2V_k - V_m}{X_{km}} \tag{3.15}$$

- Voltage angle mainly influences the active power.
- Voltage magnitude mainly influences the reactive power.

[2 mark]

ii) To simplify the power flow analysis, the so-called DC power flow approximation can be used, where the relationship between the active power and the bus angle difference is described in a linear manner. In this case, we can assume that $V_k \approx V_m \approx 1$p.u. and $\sin\theta_{km} \approx \theta_{km}$. Rewrite the expression of P_{km} as a function of θ_{km}.

Similar to $\mathbf{I} = Y\mathbf{V}$ as derived in a), which describes the relationship between node current injection and the bus voltage through the admittance matrix, the relationship between the node active power injection and the bus voltage angle can also be describe in a linear fashion through the B matrix in the DC power flow approximation. Derive the expression the matrix B as in $\mathbf{P} = B\boldsymbol{\theta}$, where $\mathbf{P}$ is the vector of bus active power injection P_k and $\boldsymbol{\theta}$ is the vector of bus voltage angle θ_k. Formulate matrix B as a function of line reactance X_{km}.

$$P_{km} = \frac{\theta_{km}}{X_{km}} = \frac{\theta_k - \theta_m}{X_{km}} \tag{3.16}$$

[1 mark]

$$P_1 = \sum_{m\in\Omega_1} \frac{\theta_1 - \theta_m}{X_{1m}} \tag{3.17}$$

$$P_2 = \sum_{m\in\Omega_2} \frac{\theta_2 - \theta_m}{X_{2m}} \tag{3.18}$$

$$\ldots \tag{3.19}$$

$$P_k = \sum_{m\in\Omega_k} \frac{\theta_k - \theta_m}{X_{km}} \tag{3.20}$$

where $m \in \Omega_k$ is the set of buses connected to bus k (not including k).

[2 marks]

$$B_{km} = -\frac{1}{X_{km}} \tag{3.21}$$

$$B_{kk} = \sum_{m\in\Omega_k} \frac{1}{X_{km}} \tag{3.22}$$

[2 marks]

4. **Question on balanced fault analysis**

a) i)

- The fault response of a synchronous generator is dictated by its sub-transient reactance and the circuit impedance up to the fault point. The fault current contribution could be as high as 5-10 p.u.
- The fault response of an IBR is dictated by its control system typically injecting reactive current based on terminal voltage. Fault current contribution is typically limited to 1.5 p.u.

[1 mark each - 2 marks]

ii) It is reasonable to neglect the pre-fault line currents because

- Line currents during fault are typically much larger than the pre-fault line currents.
- Line currents during fault are predominantly reactive while the pre-fault line currents are predominantly resistive which leads to a relatively small error if the latter is neglected.

[1 mark each - 2 marks]

iii) For a closed circuit with resistance R and emf e

$$\sum \left(Ri + \frac{d\psi}{dt} \right) = \sum e$$

Integrating over a finite time interval Δt,

$$\sum \left(R \int_0^{\Delta t} i\, dt \right) + \Delta \psi = \sum \int_0^{\Delta t} e\, dt$$

For finite R and e, as $\Delta t \to 0$, $\Delta \psi \to 0$.

[1 mark each - 2 marks]

iv)

- On a windy and sunny day very few synchronous generators are connected with inverter-based resources (IBRs) operating close to capacity reducing fault level due to the limited overcurrent capability of the IBRs.
- On a day without much wind, or sun, many IBRs stay connected at low loading alongside synchronous generators which could increase fault level.

[1 mark each - 2 marks]

b) i) Change in bus voltage, ΔV due to the fault:

$$\Delta V = Z_{\text{BUS}} J_f$$

where

$$J_f = \begin{bmatrix} 0 & 0 & \cdots & I_k (= -I_f) & \cdots 0 \end{bmatrix}^T$$

The voltage at the faulted bus V_k^f is:

$$V_k^f = V_k^0 - Z_{kk} I_f = Z_f I_f$$

The fault current I_f is calculated as:

$$I_f = \frac{V_k^0}{Z_{kk} + Z_f}$$

[1 mark each - 4 marks]

ii) Voltage at any other bus

$$\begin{aligned} V_b^f &= V_b^0 - Z_{ik} I_f \\ &= V_b^0 - \frac{Z_{bk}}{Z_{kk} + Z_f} V_k^0, \quad i = 1, 2, \ldots, N_b \end{aligned}$$

[1 mark each - 2 marks]

c) i) Faulted bus $k = 3$

Fault current is

$$I_f = \frac{V_3^0}{Z_{33} + Z_f} = \frac{1.0}{j0.22} = -j4.55 \text{ p.u.}$$

$$|I_f| = 4.55 \text{ p.u.}$$

[2 marks]

ii) Voltages at bus 1, 2, and 4 during fault are:

$$|V_1^f| = |V_1^0 - Z_{13} I_f| = 0.77 \text{ p.u.}$$

$$|V_2^f| = |V_2^0 - Z_{23} I_f| = 0.82 \text{ p.u.}$$

$$|V_3^f| = |V_3^0 - Z_{43} I_f| = 0.95 \text{ p.u.}$$

[1 mark each for correct expressions and answers - 2 marks]

iii) The Thevenin equivalent impedance of the power system at a particular bus is given by the corresponding diagonal element of Z_b

$$Z_{\text{Th-2}} = Z_{22} = j0.25 \text{ p.u.}$$

$$Z_{\text{Th-4}} = Z_{44} = j0.18 \text{ p.u.}$$

[1 mark each - 2 marks]

5. **Question on unbalanced fault analysis**

a) Fault current and voltage in phase domain:

$$I_B = I_C = 0, \quad V_A = I_A Z_f$$

[1 mark]

Fault Current and Voltage in Sequence Domain:

$$\begin{bmatrix} I_{A0} \\ I_{A1} \\ I_{A2} \end{bmatrix} = \frac{1}{3} \begin{bmatrix} 1 & 1 & 1 \\ 1 & \alpha & \alpha^2 \\ 1 & \alpha^2 & \alpha \end{bmatrix} \begin{bmatrix} I_A \\ 0 \\ 0 \end{bmatrix} = \frac{1}{3} \begin{bmatrix} I_A \\ I_A \\ I_A \end{bmatrix}$$

[1 mark]

$$I_{A0} = I_{A1} = I_{A2} = \frac{1}{3} I_A$$

[1 mark]

$$V_A = V_{A0} + V_{A1} + V_{A2} = I_A Z_f = 3 I_{A0} Z_f$$

[1 mark]

Thus, for an LG fault, the positive, negative, and zero sequence networks are connected in series with 3 times the fault impedance.

[1 mark]

[Total - 5 marks]

b) Given: $I_A = 0, \quad I_C = -I_B = 580\,\text{A}$

No zero sequence components as the sum of three phase currents is zero

$$I_{A0} = I_{B0} = I_{C0} = 0$$

[1 mark]

Positive (A-B-C) sequence component

$$\begin{aligned} I_{A1} &= \frac{1}{3}(\alpha^2 - \alpha) I_C = -j \frac{580}{\sqrt{3}}\,\text{A} = 334.86 \angle -90^\circ\,\text{A} \\ I_{B1} &= I_{A1} \times \alpha^2 = 334.86 \angle -150^\circ\,\text{A} \\ I_{C1} &= I_{A1} \times \alpha = 334.86 \angle 30^\circ\,\text{A} \end{aligned}$$

[2 marks]

Negative sequence component

$$\begin{aligned} I_{A2} &= \frac{1}{3}(\alpha - \alpha^2) I_B = j \frac{580}{\sqrt{3}}\,\text{A} = 334.86 \angle 90^\circ\,\text{A} \\ I_{B2} &= I_{A2} \times \alpha = 334.86 \angle -150^\circ\,\text{A} \\ I_{C2} &= I_{A2} \times \alpha^2 = 334.86 \angle -30^\circ\,\text{A} \end{aligned}$$

[2 marks]

[Total - 5 marks]

c) i) Equivalent sequence impedance at fault point:

$$Z_{1eq} = j(X_{G1} + X_{T1} + X_{L1}) = j0.16\ \text{p.u.}$$

$$Z_{2eq} = j(X_{G2} + X_{T2} + X_{L2}) = j0.16\ \text{p.u.}$$

The generator won't appear in the zero-sequence network due to the delta connection of the transformer winding on the generator side.

$$Z_{0eq} = j(X_{T0} + X_{L0}) = j0.27 \text{ p.u.}$$

[1 mark each – 3 marks]

Fault current

$$I_f = \frac{3}{Z_{1eq} + Z_{2eq} + Z_{0eq}} = -j5.08 \text{ p.u.}$$

[1 mark]

Base current

$$I_b = \frac{S_b}{\sqrt{3}V_b} = \frac{100}{\sqrt{3} \times 66} = 0.87 \text{ kA}$$

[1 mark]

Fault current

$$|I_f| \times I_b = 4.45 \text{ kA}$$

[1 mark]
[Total - 6 marks]

ii) Voltage on phase B, V_b

$$\begin{aligned} V_b &= V_{a0} + \alpha^2 V_{a1} + \alpha V_{a2} \\ &= \left(-\frac{Z_{0eq} \times I_f}{3}\right) + \alpha\left(-jZ_{2eq} \times \frac{I_f}{3}\right) + \alpha^2\left(1 - jZ_{1eq} \times \frac{I_f}{3}\right) \\ &= \alpha^2 - \frac{I_f}{3}\left(\alpha^2 \times Z_{1eq} + \alpha \times Z_{2eq} + Z_{0eq}\right) \\ &= -0.6864 - j0.8660 = 1.1051\angle -128.4^\circ \text{ p.u.} \end{aligned}$$

[1 mark for each step – 3 marks]

Voltage magnitude

$$|V_b| = 1.1051 \times 66 = 72.94 \text{ kV}$$

[1 mark]
[Total - 4 marks]

6. **Question on stability analysis**

a) Transient stability is the ability of a power system to maintain synchronism when subjected to a severe disturbance like faults on transmission lines, transformers, buses; loss of generation and/or loads.

[2 marks]

b) Transient stability depends on (1) the initial operating condition and (2) the severity of the disturbance.

[1 mark]

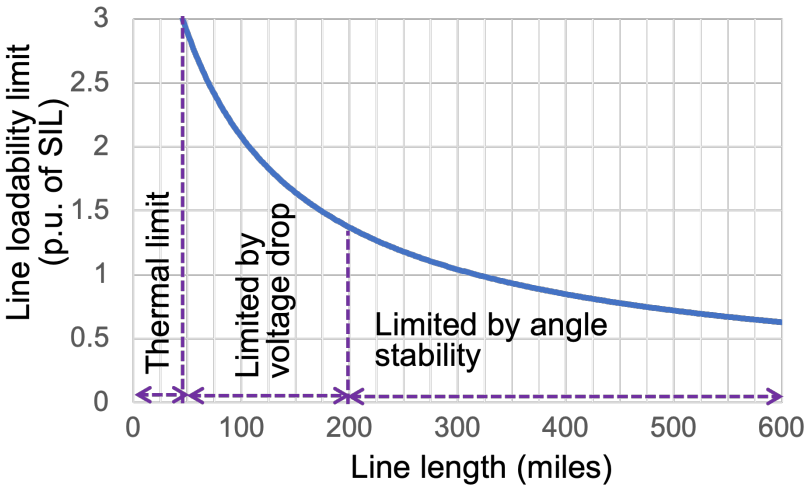
An example of an initial operating condition is the initial power angle (δ_0) or power transfer level (P_{m0}) and that of the severity of the disturbance is the magnitude of the fault current and/or duration of the fault.

[2 marks]

[Total - 3 marks]

c)
- For short distances the transmission capacity is limited by the thermal limit of the conductors.
- For medium lengths, the transmission capacity is limited by voltage drop or voltage stability.
- For longer lines the transmission capacity is limited by angle stability.

[1 mark each - 3 marks]



[2 marks]

[Total - 5 marks]

d) The two factors that limit series compensation level are: 1) network protection and 2) sub-synchronous resonance between a thermal power plant with a multi-stage turbine shaft and a radial series compensated line.

[1 mark each - 2 marks]

e) i) The total reactance between the generator internal voltage and large power grid

$$X_{is} = X_G + X_T = 0.6 \text{ p.u.}$$

$$P_{\max} = \frac{E_i \times V_s}{X_{is}} = 1.67 \text{ p.u.}$$

[1 mark]

Initial power angle

$$\delta_0 = \sin^{-1}\left(\frac{P \times X_{is}}{E_i \times V_s}\right) = 48.59^\circ = 0.85\text{rad}$$

[1 mark]

Using the Equal Area Criterion

$$\begin{aligned}\int_{\delta_0}^{\delta_{cr}} (P-0)\,d\delta &= \int_{\delta_{cr}}^{\pi-\delta_0} (P_{\max}\sin\delta - P)\,d\delta \\ P(\delta_{cr}-\delta_0) &= P_{\max}(\cos\delta_{cr} - \cos(\pi-\delta_0)) - P(\pi-\delta_0-\delta_{cr}) \\ \delta_{cr} &= \cos^{-1}[(\pi-2\delta_0)\sin\delta_0 - \cos\delta_0] = 1.13 \text{ rad} = 65^\circ\end{aligned}$$

[2 marks]

Critical clearing time

$$t_{cr} = \sqrt{\frac{4H(\delta_{cr}-\delta_0)}{2\pi f \times P}} = 0.15 \text{ sec}$$

[1 mark]
[Total - 5 marks]

ii) With series compensation, the total reactance between the generator internal voltage and large power grid is

$$X_{is1} = X_G + X_T(1-k) = 0.42 \text{ p.u.}$$

Initial power angle

$$\delta_1 = \sin^{-1}\left(\frac{P \times X_{is1}}{E_i \times V_s}\right) = 31.67^\circ = 0.55 \text{ rad}$$

[1 mark]

New critical clearing angle

$$\delta_{cr1} = \cos^{-1}[(\pi-2\delta_1)\sin\delta_1 - \cos\delta_1] = 1.35 \text{ rad} = 77.42^\circ$$

[1 mark]

New critical clearing time

$$t_{cr1} = \sqrt{\frac{4H(\delta_{cr1}-\delta_0)}{2\pi f \times P}} = 0.20 \text{ sec}$$

[1 mark]
[Total - 3 marks]